Assignment 10: Virtual Memory and Secondary Storage

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Answer every question. Show all working for Numerical and Design questions.

10.1 Conceptual

Q1. A TLB is a small associative cache of page translations, while a page table is the larger map from VPN to PFN. Explain why the TLB must compare stored VPNs in parallel, and describe the complete path for (a) a TLB hit and (b) a TLB miss whose page-table entry is valid. Include the hardware cost that associative lookup introduces.

Q2. A memory system can use fixed-size paging or variable-size segmentation. Compare the two schemes for (a) the unit being placed, (b) the kind of fragmentation each makes likely, and (c) protection or sharing. Then explain why a page fault is not automatically a program error, while an illegal address or permission failure may terminate the process.

10.2 Numerical

Q3. A system has a 20-bit virtual address, 1 KiB pages, and 16 physical frames. The page-table entry for the VPN of virtual address `0xABCDE` is valid and maps that page to physical frame 9.

(a) Compute the virtual-page-number and offset field widths.

(b) Split `0xABCDE` into VPN and offset, giving both hexadecimal and decimal values.

(c) Compute the physical-address field widths and the translated physical address in hexadecimal and decimal. Verify that the offset is unchanged.

Q4. A disk access has seek time 8 ms, rotational latency 3 ms, transfer time 1.5 ms, and controller time 0.5 ms. A RAID 5 array stores three corresponding four-bit data blocks

$$D_0 = 1101, \qquad D_1 = 0011, \qquad D_2 = 0101.$$

(a) Compute the HDD access time using the lecture's decomposition.

(b) Compute the RAID 5 parity block $P = D_0 \oplus D_1 \oplus D_2$.

(c) If D_2 fails, reconstruct it from D_0, D_1, and P, showing both XOR steps.

10.3 Design

Q5. A process references a virtual page whose page-table entry is invalid. There is no free physical frame, and the selected victim page is dirty. Design the page-fault handling sequence from the hardware trap through restart of the faulting instruction. State where the validity/protection check, dirty write-back, page-in, page-table update, and TLB update occur. Then state how the sequence changes if the access fails the protection check.

Q6. A storage-backed system must support two workloads: (i) an active operating-system working set with frequent random accesses and (ii) a server workload that needs high throughput and continued availability after one disk failure. For each workload, choose the most defensible medium or organisation from the lecture's table (SSD/flash, HDD, optical, RAID 0, RAID 1, or RAID 5), and justify the choice using the relevant bottleneck and trade-off. Explain why the other choices are less suitable for that workload.

References